

DiS Chapter 4 Notes

Functional Units of Digital Systems

4.1 Von Neumann architecture, Functional units of Computer, Basic Instruction Cycle, Interrupts

4.2 Memory sub-system-Typical architecture of RAM and ROM, SRAM and DRAM – architecture and comparison, Memory hierarchy

Von-Neumann computer architecture design was proposed in 1945. It was later known as Von-Neumann architecture.

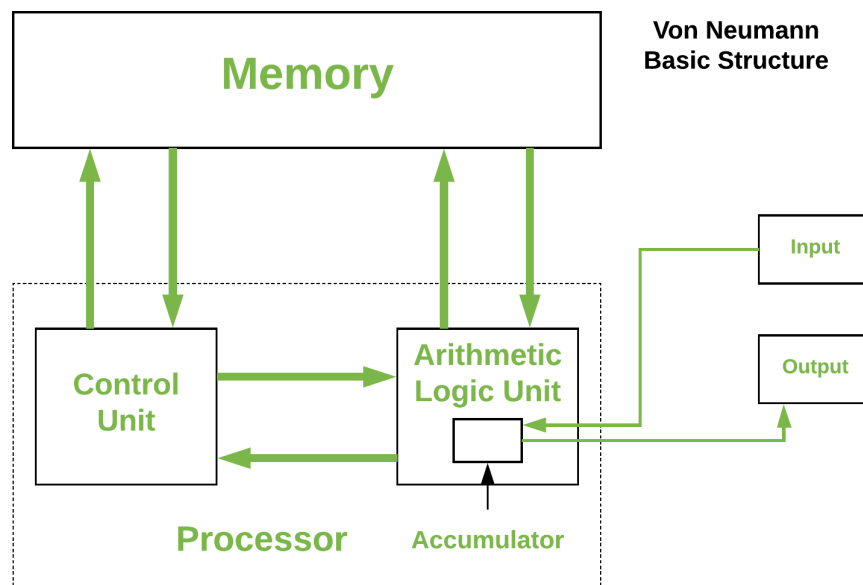
Historically there have been 2 types of Computers:

Fixed Program Computers: Their function is very specific and they couldn't be reprogrammed, e.g. Calculators.

Stored Program Computers: These can be programmed to carry out many different tasks, applications are stored on them, hence the name.

Modern computers are based on a stored-program concept introduced by John Von Neumann. In this stored-program concept, programs and data are stored in a separate storage unit called memories and are treated the same. This novel idea meant that a computer built with this architecture would be much easier to reprogram.

The basic structure is like this,



It is also known as ISA (Instruction set architecture) computer and is having three basic units:

- The Central Processing Unit (CPU)
- The Main Memory Unit
- The Input/Output Device Let's consider them in detail.

Central Processing Unit (CPU) : Once the information is entered into the computer by the input device, the processor processes it. The CPU is called the brain of the computer because it is the control center of the computer. It first fetches instructions from memory and then interprets them so as to know what is to be done. If required, data is fetched from memory or input device. Thereafter the CPU executes or performs the required computation and then either stores the output or displays on the output device. The CPU has three main components which are responsible for different functions – Arithmetic Logic Unit (ALU), Control Unit (CU) and Memory registers

Control Unit: The Control unit coordinates and controls the data flow in and out of CPU and also controls all the operations of ALU, memory registers and also input/output units. It is also responsible for carrying out all the instructions stored in the program. It decodes the fetched instruction, interprets it and sends control signals to input/output devices until the required operation is done properly by ALU and memory.

Arithmetic and Logic Unit (ALU): The arithmetic logic unit is that part of the CPU that handles all the calculations the CPU may need, e.g. Addition, Subtraction, Comparisons. It performs Logical Operations, Bit Shifting Operations, and Arithmetic operations.

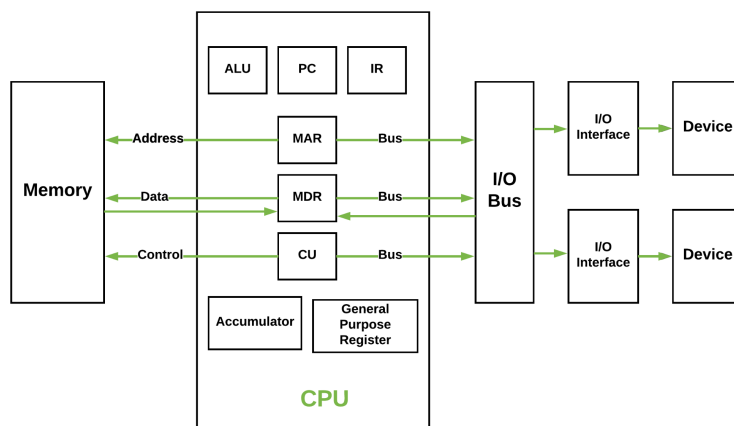


Figure – Basic CPU structure, illustrating ALU

Memory: Memory attached to the CPU is used for storage of data and instructions and is called internal memory. The internal memory is divided into many storage locations, each of which can store data or instructions. Each memory location is of the same size and has an address. With the help of the address, the computer can read any memory location easily without having to search the entire memory. When a program is executed, its data is copied to the internal memory and is stored in the memory till the end of the execution.

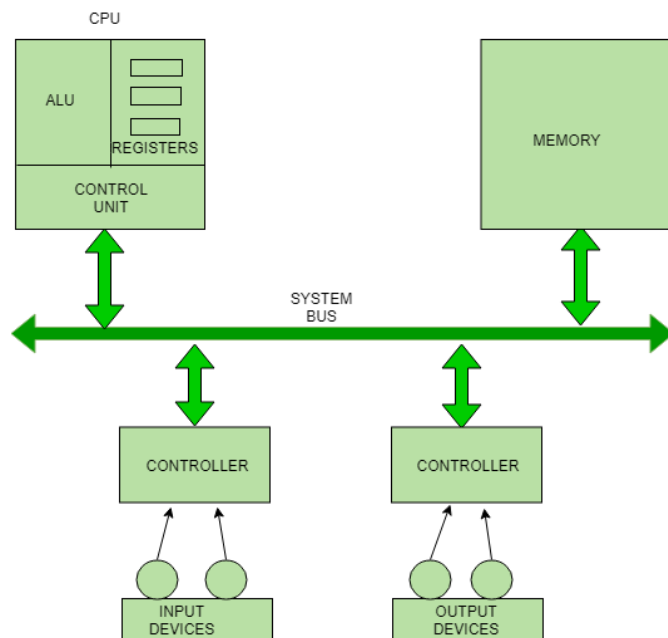
Registers: Registers refer to high-speed storage areas in the CPU. The data processed by the CPU are fetched from the registers. There are different types of registers used in architecture:-

- **Accumulator:** Stores the results of calculations made by ALU. It holds the intermediate of arithmetic and logical operations. It acts as a temporary storage location or device.
- **Program Counter (PC):** Keeps track of the memory location of the next instructions to be dealt with. The PC then passes this next address to the Memory Address Register (MAR).
- **Memory Address Register (MAR):** It stores the memory locations of instructions that need to be fetched from memory or stored in memory.
- **Memory Data Register (MDR):** It stores instructions fetched from memory or any data that is to be transferred to, and stored in, memory.
- **Current Instruction Register (CIR):** It stores the most recently fetched instructions while it is waiting to be coded and executed.
- **Instruction Buffer Register (IBR):** The instruction that is not to be executed immediately is placed in the instruction buffer register IBR.

Buses: Data is transmitted from one part of a computer to another, connecting all major internal components to the CPU and memory, by the means of Buses. Types:

- **Data Bus:** It carries data among the memory unit, the I/O devices, and the processor.
- **Address Bus:** It carries the address of data (not the actual data) between memory and processor.
- **Control Bus:** It carries control commands from the CPU (and status signals from other devices) in order to control and coordinate all the activities within the computer.

Input/Output Devices: Program or data is read into main memory from the input device or secondary storage under the control of CPU input instruction. Output devices are used to output information from a computer. If some results are evaluated by the computer and it is stored in the computer, then with the help of output devices, we can present them to the user.

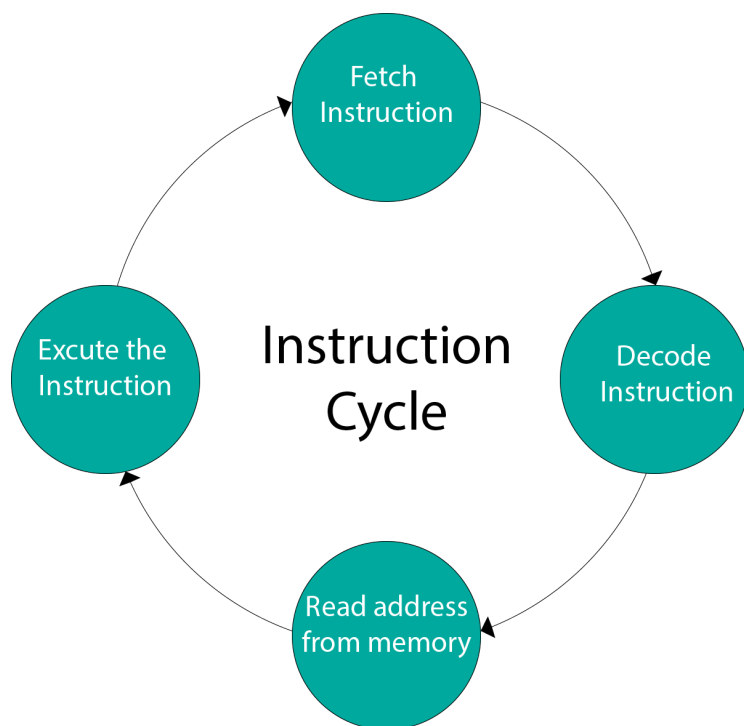


Von Neumann bottleneck:

Whatever we do to enhance performance, we cannot get away from the fact that instructions can only be done one at a time and can only be carried out sequentially. Both of these factors hold back the competence of the CPU. This is commonly referred to as the 'Von Neumann bottleneck'. We can provide a Von Neumann processor with more cache, more RAM, or faster components but if original gains are to be made in CPU performance then an influential inspection needs to take place of CPU configuration. This architecture is very important and is used in our PCs and even in Supercomputers.

Instruction Cycle: A program residing in the memory unit of a computer consists of a sequence of instructions. These instructions are executed by the processor by going through a cycle for each instruction. In a basic computer, each instruction cycle consists of the following phases:

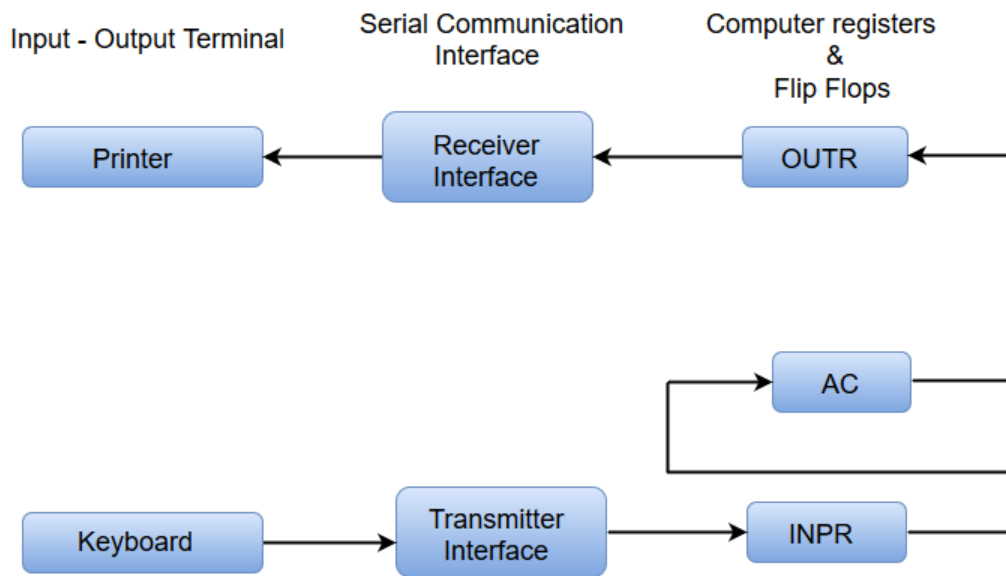
- Fetch instruction from memory.
- Decode the instruction.
- Read the effective address from memory.
- Execute the instruction.



Input-Output Configuration: In computer architecture, input-output devices act as an interface between the machine and the user.

Instructions and data stored in the memory must come from some input device. The results are displayed to the user through some output device.

Input - Output Configuration:



- The input-output terminals send and receive information.
- The amount of information transferred will always have eight bits of an alphanumeric code.
- The information generated through the keyboard is shifted into an input register 'INPR'.
- The information for the printer is stored in the output register 'OUTR'.
- Registers INPR and OUTR communicate with a communication interface serially and with the AC in parallel.
- The transmitter interface receives information from the keyboard and transmits it to INPR.
- The receiver interface receives information from OUTR and sends it to the printer serially.

Interrupts: The interrupt is a signal emitted by hardware or software when a process or an event needs immediate attention. It alerts the processor to a high-priority process requiring interruption of the current working process. In I/O devices one of the bus control lines is dedicated for this purpose and is called the Interrupt Service Routine (ISR). When a device raises an interrupt at let's say process i , the processor first completes the execution of instruction i . Then it loads the Program Counter (PC) with the address of the first instruction of the ISR. Before loading the Program Counter with the address, the address of the interrupted instruction is moved to a temporary location. Therefore, after handling the interrupt the processor can continue with process $i+1$. While the processor is handling the interrupts, it must inform the device that its request has been recognized so that it stops sending the interrupt request signal. Also, saving the registers so that the interrupted process can be restored in the future, increases the delay between the time an interrupt is received and the start of the execution of the ISR. This is called Interrupt Latency.

Software Interrupts: A sort of interrupt called a software interrupt is one that is produced by software or a system as opposed to hardware. Traps and exceptions are other names for software interruptions. They serve as a signal for the operating system or a system service to carry out a certain function or respond to an error condition. A particular instruction known as a “interrupt instruction” is used to create software interrupts. When the interrupt instruction is used, the processor stops what it is doing and switches over to a particular interrupt handler code. The interrupt handler routine completes the required work or handles any errors before handing back control to the interrupted application.

Hardware Interrupts: In a hardware interrupt, all the devices are connected to the Interrupt Request Line. A single request line is used for all the n devices. To request an interrupt, a device closes its associated switch. When a device requests an interrupt, the value of INTR is the logical OR of the requests from individual devices.

The sequence of events involved in handling an IRQ:

- Devices raise an IRQ.
- The processor interrupts the program currently being executed.
- The device is informed that its request has been recognized and the device deactivates the request signal.
- The requested action is performed.
- An interrupt is enabled and the interrupted program is resumed.

Handling Multiple Devices:

When more than one device raises an interrupt request signal, then additional information is needed to decide which device to be considered first. The following methods are used to decide which device to select: Polling, Vectored Interrupts, and Interrupt Nesting. These are explained below.

- Polling: In polling, the first device encountered with the IRQ bit set is the device that is to be serviced first. Appropriate ISR is called to service the same. It is easy to implement but a lot of time is wasted by interrogating the IRQ bit of all devices.
- Vectored Interrupts: In vectored interrupts, a device requesting an interrupt identifies itself directly by sending a special code to the processor over the bus. This enables the processor to identify the device that generated the interrupt. The special code can be the starting address of the ISR or where the ISR is located in memory and is called the interrupt vector.
- Interrupt Nesting: In this method, the I/O device is organized in a priority structure. Therefore, an interrupt request from a higher priority device is recognized whereas a request from a lower priority device is not. The processor accepts interrupts only from devices/processes having priority.

Processors' priority is encoded in a few bits of PS (Process Status register). It can be changed by program instructions that write into the PS. The processor is in supervised mode only while executing OS routines. It switches to user mode before executing application programs.

Memory is the most essential element of a computing system because without it computer can't perform simple tasks. Both types of memory (RAM and ROM) are important for the computer, but they serve different purposes. RAM is used to store data that the computer is currently using, while ROM is used to store data that the computer needs to boot and operate. RAM is faster than ROM, as the data stored in it can be accessed and modified in any order, while data stored in ROM can only be read.

Random Access Memory (RAM) is a type of computer memory that is used to temporarily store data that the computer is currently using or processing. RAM is volatile memory, which means that the data stored in it is lost when the power is turned off. RAM is typically used to store the operating system, application programs, and data that the computer is currently using.

Read Only Memory (ROM) is a type of computer memory that is used to permanently store data that does not need to be modified. ROM is non-volatile memory, which means that the data stored in it is retained even when the power is turned off. ROM is typically used to store the computer's BIOS (basic input/output system), which contains the instructions for booting the computer, as well as firmware for other hardware devices.

Static RAM: SRAM stands for Static Random Access Memory. It is a type of semiconductor which is widely used in computing devices and microprocessors.

Dynamic RAM: DRAM stands for Dynamic Random Access Memory. It is made of Capacitors and has a smaller data lifespan than Static RAM.

DRAM	SRAM
1. Constructed of tiny capacitors that leak electricity.	1.Constructed of circuits similar to D flip-flops.
2.Requires a recharge every few milliseconds to maintain its data.	2.Holds its contents as long as power is available.
3.Inexpensive.	3.Expensive.
4. Slower than SRAM.	4. Faster than DRAM.
5. Can store many bits per chip.	5. Can not store many bits per chip.
6. Uses less power.	6.Uses more power.
7.Generates less heat.	7.Generates more heat.
8. Used for main memory.	8. Used for cache.

Difference between SRAM and DRAM

Advantages of Random Access Memory (RAM)

- **Speed:** RAM is much faster than other types of storage, such as a hard drive or solid-state drive, which means that the computer can access the data stored in RAM more quickly.
- **Flexibility:** RAM is volatile memory, which means that the data stored in it can be easily modified or deleted. This makes it ideal for storing data that the computer is currently using or processing.
- **Capacity:** The capacity of RAM can be easily upgraded, which allows the computer to store more data in memory and thus improve performance.
- **Power Management:** RAM consumes less power compared to hard drives, and solid-state drives, which makes it an ideal memory for portable devices.

Disadvantages of Random Access Memory (RAM)

- **Volatility:** RAM is volatile memory, which means that the data stored in it is lost when the power is turned off. This can be a problem for important data that needs to be preserved, such as unsaved work or files that have not been backed up.
- **Capacity:** The capacity of RAM is limited, and although it can be upgraded, it may still not be sufficient for certain applications or tasks that require a lot of memory.
- **Cost:** RAM can be relatively expensive compared to other types of memory, such as hard drives or solid-state drives, which can make upgrading the memory of a computer or device more costly.

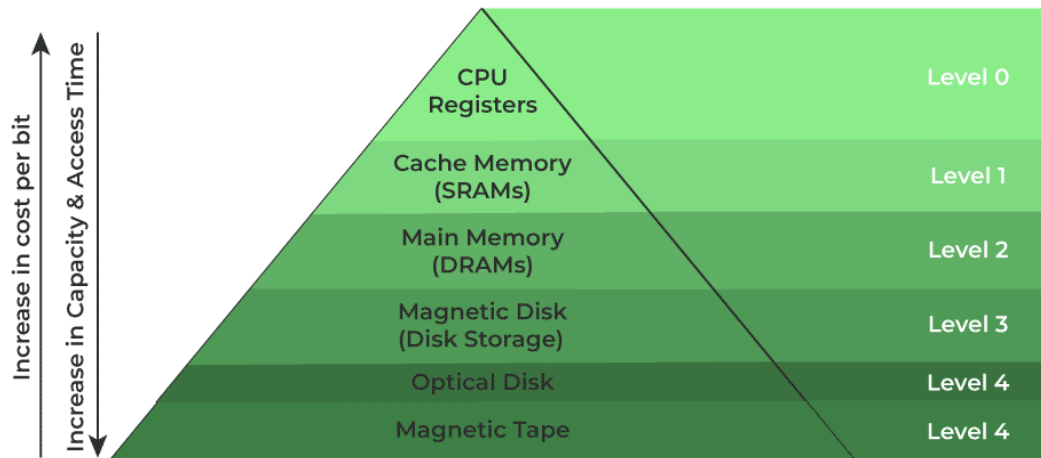
Read-Only Memory (ROM)

- Stores crucial information essential to operate the system, like the program essential to boot the computer.
- It is non-volatile.
- Always retains its data.
- Used in embedded systems or where the programming needs no change.
- Used in calculators and peripheral devices.
- ROM is further classified into four types- MROM, PROM, EPROM, EEPROM

Memory Hierarchy is one of the most required things in Computer Memory as it helps in optimizing the memory available in the computer. There are multiple levels present in the memory, each one having a different size, different cost, etc. Accessing data is not similar in all types of memory, some have faster access whereas some have slower access.

Types of Memory Hierarchy

- **External Memory or Secondary Memory:** Comprising Magnetic Disk, Optical Disk, and Magnetic Tape i.e. peripheral storage devices which are accessible by the processor via an I/O Module.
- **Internal Memory or Primary Memory:** Comprising Main Memory, Cache Memory & CPU registers. This is directly accessible by the processor.



Memory Hierarchy Design

Memory Hierarchy Design:

- **Registers:** Registers are small, high-speed memory units located in the CPU. They are used to store the most frequently used data and instructions. Registers have the fastest access time and the smallest storage capacity, typically ranging from 16 to 64 bits.
- **Cache Memory:** Cache memory is a small, fast memory unit located close to the CPU. It stores frequently used data and instructions that have been recently accessed from the main memory. Cache memory is designed to minimize the time it takes to access data by providing the CPU with quick access to frequently used data.
- **Main Memory:** Main memory, also known as RAM (Random Access Memory), is the primary memory of a computer system. It has a larger storage capacity than cache memory, but it is slower. Main memory is used to store data and instructions that are currently in use by the CPU. Types of Main Memory: SRAM and DRAM.
- **Secondary Storage:** Secondary storage, such as hard disk drives (HDD) and solid-state drives (SSD), is a non-volatile memory unit that has a larger storage capacity than main memory. It is used to store data and instructions that are not currently in use by the CPU. Secondary storage has the slowest access time and is typically the least expensive type of memory in the memory hierarchy.
- **Magnetic Disk:** Magnetic Disks are simply circular plates that are fabricated with either a metal or a plastic or a magnetized material. The Magnetic disks work at a high speed inside the computer and these are frequently used.
- **Magnetic Tape:** Magnetic Tape is simply a magnetic recording device that is covered with a plastic film. It is generally used for the backup of data. In the case of a magnetic tape, the access time for a computer is a little slower and therefore, it requires some amount of time for accessing the strip.

Characteristics of Memory Hierarchy

- Capacity: It is the global volume of information the memory can store. As we move from top to bottom in the Hierarchy, the capacity increases.
- Access Time: It is the time interval between the read/write request and the availability of the data. As we move from top to bottom in the Hierarchy, the access time increases.
- Performance: Earlier when the computer system was designed without a Memory Hierarchy design, the speed gap increased between the CPU registers and Main Memory due to a large difference in access time. This results in lower performance of the system and thus, enhancement was required. This enhancement was made in the form of Memory Hierarchy Design because of which the performance of the system increases. One of the most significant ways to increase system performance is minimizing how far down the memory hierarchy one has to go to manipulate data.
- Cost Per Bit: As we move from bottom to top in the Hierarchy, the cost per bit increases i.e. Internal Memory is costlier than External Memory.

Advantages of Memory Hierarchy

- It helps in removing some destruction, and managing the memory in a better way.
- It helps in spreading the data all over the computer system.
- It saves the consumer's price and time.